ANALYTICAL CUMULATIVE EXAMINATION

24th April 1993

The compositions of the surfaces of planetary bodies can be analyzed using the energetic radiations they produce. To date, artificial satellites carrying X-ray and gamma-ray spectrometers have analyzed the global composition of the lunar surface, and satellites have been designed to study the surfaces of Mars, the asteroids and comets; one is currently on its way to Mars, another is likely to be flown to an asteroid in the near future.

1. (a) Describe the design of a spectrometer suitable for the analysis of the surface of a planetary body using X-rays.

(b) Describe the mechanism by which the X-rays are produced, emphasizing the relevant laws, and describe how the concentrations are determined from the spectra.

(c) Describe a comparable laboratory instrument and list some of the differences and similarities to that flown on spacecraft.

2. (a) Describe the design of a spectrometer suitable for the analysis of the surface of a planetary body using gamma radiation.

(b) Describe the mechanism by which the gamma radiation is produced, and describe how concentrations may be determined from the spectra.

(c) Describe a comparable means of analysis in the laboratory and list and explain some of the differences and similarities to that used on spacecraft.

3. Describe briefly some of the main results of such measurements of the lunar surface.

The first two questions carry 45% weighting each, the third is worth 10%. Special credit will be given for accurate figures and quantitative answers.

		15	P 30.97376 $\hat{\sigma}_a$ .180, .08	P 27 26.9994	P 28 3+ 270ms β^+ 11.5, ··· γ 1.779, E 14.33	P 29 1/+ 4.14s β^+ 3.94, γ 1.27, 2.43 E 4.94	P 30 1+ 2.50m β^+ 3.24, , ϵ γ 2.235 E 4.227	
14	Si 28.0855 $\bar{\sigma}_a$ .16, .08		Si 25 220ms β^+ (p) 4.43, 2.03, 3.61, ··· E 16.74 25.00411	Si 26 2.20s β^+ 3.815, ··· γ .829, ··· E 5.064	Si 27 5/+ 4.14s β^+ 3.8, ··· γ 2.21, ··· E 4.809	Si 28 92.23 $\bar{\sigma}_\gamma$ .16, .08 27.976928	Si 29 1/+ 4.67 $\bar{\sigma}_\gamma$ .10, .07 28.976496	
Al 26.98154 σ_a .233, .17		Al 23 (5/+) 0.47s β^+ (p) .92 E 12.24 23.00727	Al 24 4+ 0.129s IT .439 β^+ 13.3, 11.9, ···, γ 1.37, ···, (α) 1.40, 1.77, ··· 2.07s β^+ 3.4, 4.4, ···, γ 1.37, ···, (α) 1.56-2.36 E 13.88	Al 25 5/+ 7.23s β^+ 3.26, ··· γ 1.611, ··· E 4.278	Al 26 5+ 6.34s β^+ 3.21 7.3x10^{5}a β^+ 1.17, ϵ, γ 1.809, ··· E 4.005	Al 27 5/+ 100 σ_γ .233, .17 26.981541	Al 28 3+ 2.24m β^- 2.87 γ 1.779 E 4.643	
Mg 20? 0.6s β^+ (p) E 10.73 20.01886	Mg 21 (5/)+ 0.122s β^+ (p) .95-6.54 E 13.10 21.01172	Mg 22 3.86s β^+ γ .583, .074, ··· E 4.790 21.999577	Mg 23 3/+ 11.3s β^+ 3.1, ··· γ .439 E 4.059	Mg 24 78.99 $\hat{\sigma}_\gamma$ .053, .032 23.985045	Mg 25 5/+ 10.00 $\hat{\sigma}_\gamma$ .17, .10 24.985839	Mg 26 11.01 σ_γ .036, .025 25.982595	Mg 27 1/+ 9.45m β^- 1.75, 1.59 γ .8438, 1.0144, ··· σ_γ .15 E 2.609	
Na 19 0.03s? β^+ (p) E 11.18 19.01388	Na 20 2+ 0.446s β^+ 11.24, ··· γ 1.633, ··· (α) 2.15, ··· E 13.89	Na 21 3/+ 22.5s β^+ 2.51, ··· γ .35 E 3.547	Na 22 3+ 2.601a β^+ .545, ···, ϵ γ 1.2745 σ_γ 4x10^{-3} 1.6x10^{-3} E 2.842	Na 23 3/+ 100 σ_γ (.40+.13), .32 22.989770	Na 24 4+ 20.0ms IT .4723 β^- ~6 15.02h β^- 1.390, ···, γ 2.754, 1.369 E 5.513	Na 25 5/+ 60s β^- 3.8, ··· γ .975, .585, .390, ··· E 3.83	Na 26 (3+) 1.07s β^- 7.4, ··· γ 1.809 E 9.33	
Ne 18 1.67s β^+ 3.42, 2.38, ··· γ 2, ··· E 4.45	Ne 19 1/+ 17.2s β^+ 2.22, ··· γ 1.357, ··· E 3.238	Ne 20 90.51 $\hat{\sigma}_\gamma$ .037, .02 19.992439	Ne 21 3/+ 0.27 $\hat{\sigma}_\alpha$ < 1.5 $\hat{\sigma}_\gamma$ .7, .3 20.993845	Ne 22 9.22 $\hat{\sigma}_\gamma$ .048, .02 21.991384	Ne 23 5/+ 37.5s β^- 4.38, 3.95, ··· γ .438, ··· E 4.375	Ne 24 3.38m β^- 1.98, ··· γ 472D, ··· E 2.47	Ne 25 0.61s β^- 7.3, ··· γ .0895, ··· E 7.2	
F 17 5/+ 64.5s β^+ 1.74 no γ E 2.762	F 18 1+ 109.8m β^+ .635, ϵ E 1.656	F 19 1/+ 100 $\hat{\sigma}_\gamma$ 9.5 mb, 20mb 18.9984033	F 20 2+ 11.0s β^- 5.40, ··· γ 1.633, ··· E 7.026	F 21 5/+ 4.33s β^- 5.34, 5.7, ··· γ .351, 1.395, ··· E 5.69	F 22 5+ 4.23s β^- 5.5, 3.5, ··· γ 1.275, 2.083, 2.166, ··· E 10.85	F 23 2.2s β^- 8.3, ··· γ 1.701, 2.129, ··· E 8.5	F 24 24.0093	
O 16 99.758 $\hat{\sigma}_\gamma$ .18mb, .26mb 15.9949146	O 17 5/+ 0.038 $\bar{\sigma}_\alpha$ .24, .11 16.999131	O 18 0.204 σ_γ .16 mb, .85mb 17.9991594	O 19 5/+ 26.8s β^- 3.25, 4.6, ··· γ .198, 1.357, ··· E 4.819	O 20 13.5s β^- 2.75 γ 1.057 E 3.82	O 21 21.0087	O 22 22.0102	O 23 23.0193	
N 15 1/- 0.37 $\hat{\sigma}_\gamma$ .02 15.00010898	N 16 2- 7.10s β^- 4.3, 10.42, ··· γ 6.129, 7.115, ··· (α) 1.28, 2.02 E 10.419	N 17 1/- 4.17s β^- 7.81, 5.62, ··· γ .871, 2.184 (n) 1.171, .383, ··· E 8.68	N 18 0.63s β^- 9.4 γ 1.98, 1.65, .82, 2.47 E 14.06	N 19 0.42s β^- γ 2.47, ··· (n) E 12.3 19.0168	N 20 20.0238	N 21 21.0289		
C 14 5730a β^- .156 no γ $\hat{\sigma}_\gamma$ < 1 μb E .15648	C 15 1/+ 2.45s β^- 4.5, 9.8, ··· γ 5.298, ···	C 16 0.75s β^- (n) .79, 1.7 E 8.01 16.01470	C 17 17.0226	C 18 18.0272	C 19 19.0370	14		
B 13 3/- 17.3ms β^- 13.4, ··· γ 3.68, ··· (n) 2.4, 3.6, ···	B 14 2- 16 ms β^- 14, ··· γ 6.09, 6.73, ···	B 15		B 17				